Midterm Examination
ECO 354: Public Economics
Department of Economics
Shiv Nadar University
Monsoon 2016.

This is a one and a half hour long closed book, closed notes in class examination. Answer all the questions, and be clear and precise in your answers. Any act of *academic dishonesty* will be taken very seriously. *All the best.*

Question 1: Pollution Externalities and Traditional Remedies (20 points)

Consider two oil refineries, both produces fuel that has a market price 3 per gallon and a per gallon cost of production 2. Denote the number of gallons firm i produces by y_i, $i \in \{1, 2\}$. Additionally, the firms also produces *smog (s)*, as a by product, and the damage function by each firm is given by: $s_1 = 0.01y_1^2$, and $s_2 = 0.01\frac{y_2^2}{2}$. Plant 2 pollutes only half as much as plant 1 does for a given level of production. The capacity of each firm is to produce 200 gallons.

(a) Assume in the beginning there is no regulation: each firm can produce as much as it can until it runs out of capacity. Set-up the optimization problem faced by each firm in such a scenario, and find out how much will each firm optimally produce in absence of any regulation? How much *smog* will be produced as a by-product of such competitive behaviour? (5 points)

(b) What would be socially efficient level of *smog* production by each firm.
Hint: Note here that the optimal level would need to equate the private benefit with the social cost and therefore, would need to *internalize the externalities.* (10 points)

(c) How does the social optimal outcome differs from the private market outcome for the two firms? Can you relate this departure with their social damage function and explain your findings. (5 points)

Question 2: Private Provision of Public Goods and Social Efficiency (30 points)

Consider two individuals with asymmetric utility functions over a composite consumption good X, and a public good G, where the utility function of individual i is given by: $u(x_i) = \alpha_i log(x_i) + (1 - \alpha_i)log(G)$, where G is a public good. Each of them is initially endowed with $w_i; i \in \{1, 2\}$. Also suppose that production technology is such that each unit of X can be converted into an unit of G, so that the economy wide MRT is given by 1.

(a) First, calculate the socially optimal provision of public good that shall be provided by a social planner as a function of α. Explain the condition that you use to get social optimality in this context. (3 points)

Consider a situation where α_i for all α

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(b) Comment how that provision depends on α_i. Precisely explain your finding. (2 points)

Now suppose the public good will have to be *privately provided*, which means that agents have to fund the contribution towards the public good and their private consumption from their initial endowment.

$\alpha_1 \neq \alpha_2$

(c) Set up the optimization problem of an agent, and find out the *contribution reaction function* of an agent i. How does that optimal strategic behaviour depend on the contribution of other agents, and the importance of public good in the preference of each agent. Briefly explain. Do you find it dependent on the income distribution? If yes, explain the intuition behind that relationship (5 points)

(d) How much each agent will contribute in a *Nash equilibrium*? How does an agent's optimal contribution depend on her own income, the income of the other agent, and the α_i's? Can you briefly comment the intuition behind each of these relationships. (6 points)

(e) How much of public good will be provided as a result? How does the provision in a NE compares with that in a social optima? Explain briefly. (4 point)

Find out **the utility that each person derives from the provision** at the *Nash* equilibrium allocation. Define the total utility derived by both the agents taken together as the *Social Benefit* from the project. Also, find out the total contribution cost, which in this context will just be the addition of individual contributions at the NE. Define a project to be socially beneficial when total benefit from the project exceeds total cost.

total utility vs cost

(f) Derive the condition under which the provision in *Nash* equilibrium is socially beneficial? How does that condition depend on the parameters of the system? Can you derive some policy implication regarding how to allocate initial endowments or, how to choose agents so that the *Nash* equilibrium outcome becomes socially beneficial. (10 points)

BONUS QUESTION (Coase Theorem) (10 points)

A Doctor and a Baker both lives in a building and have their operational units in the same building. Doctor needs quiet environment for her private practice, whereas using noisy machines is key to Baker's production and profit. Traditional wisdom would hold Baker guilty for making noise and ask her to compensate the doctor. A lawyer on Baker's side might also argue that doctor's need to have a quieter environment in fact imposes an impediment for Baker's work and therefore, Doctor should install sound-proofing. A quieter machine costs 100, whereas installation of sound-proofing is much more expensive and costs 200. While the owner of the building, hearing the plea from both sides, was contemplating of assigning some rights to one of the party, an economist, Ronald Coase, offered a solution. He asked the owner to assign the right to hassle-free work to *anyone of them*, and assured that the outcome will be *good for all*.

(a) Can you comment whether Coase was right, and if yes, can you logically explain how assigning a right to *anyone* leads to socially efficient outcome.

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Question 2.

$u(x_i) = \alpha_i \log(x_i) + (1-\alpha_i) \log G$.

where $i \in \{1, 2\}$

$w_i = x_i + g_i$

$g_1 + g_2 = G$

$MRT = 1$

taking $\alpha_1 = \alpha_2 = \alpha$.

1 (a). Socially optimal provision of public good is given when the samuelson condition holds

$$\sum_{i=0}^{n-1} MRS_i = MRT$$

$$MRS_1 = \frac{\partial u_1/\partial g_1}{\partial u_1/\partial x_1}$$

$$= \frac{\frac{1-\alpha}{g_1+g_2}}{\frac{\alpha}{x_1}} = \frac{(1-\alpha)}{(g_1+g_2)} \frac{x_1}{\alpha}$$

$$= \frac{(1-\alpha)x_1}{(g_1+g_2)\alpha}$$

Similarly $MRS_2 = \frac{\partial u_2/\partial g_2}{\partial u_2/\partial x_2} = \frac{(1-\alpha)x_2}{(g_1+g_2)\alpha}$.

$$\frac{(1-\alpha)x_1}{(g_1+g_2)\alpha} + \frac{(1-\alpha)x_2}{(g_1+g_2)\alpha} = 1.$$

$$\frac{(1-\alpha)(x_1+x_2)}{(g_1+g_2)\alpha} = 1.$$

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$\frac{(1-\alpha)}{\alpha}(x_1+x_2) = G^0$

$G^0 = \frac{(1-\alpha)}{\alpha}(x_1+x_2)$ ⇒ $(1-\alpha)\alpha^{-1}$

$\frac{\partial G^0}{\partial \alpha} =$

$\frac{(1-\alpha)}{\alpha}(w_1 - g_1 + w_2 - g_2)$

$\frac{1-\alpha}{\alpha}(w_1 + w_2 - (g_1+g_2)) = G^0$

$\frac{1-\alpha}{\alpha}(w_1+w_2) - \left(\frac{1-\alpha}{\alpha}\right)(g_1+g_2) = G^0$

$\left(\frac{1-\alpha}{\alpha}\right)(w_1+w_2) = G + \left(\frac{1-\alpha}{\alpha}\right)(g_1+g_2)$

$\left(\frac{1-\alpha}{\alpha}\right)(w_1+w_2) = G\left[1 + \frac{1-\alpha}{\alpha}\right]$

$\left(\frac{1-\alpha}{\alpha}\right)(w_1+w_2) = G\left(\frac{1}{\alpha}\right)$

$G = \left(\frac{1-\alpha}{\alpha}\right)\alpha(w_1+w_2)$

$G^0 = (1-\alpha)(w_1+w_2)$ ✓ 5

(b) $\frac{\partial G^0}{\partial \alpha} = -(w_1+w_2) < 0$ as $w_1+w_2 > 0$

As the first derivative of optimal provision of public good is negative wrt to α, it is a decreasing function. This means that as α increases G decreases.

As α is an individual valuation for the public good,

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~~So~~ a higher valuation of the public good results in a higher provision of the public good at socially optimal level.

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(c). $\text{Max } U_i(x_i) = \alpha_i \log x_i + (1-\alpha_i)\log G$

Subject $x_i + g_i = w_i$

$g_1 + g_2 = G.$

~~Ans~~

For ~~[illegible]~~ Individual 1:

$U_1(x_1) = \alpha_1 \log x_1 + (1-\alpha_1)\log G$

$U_1(x_1) = \alpha_1 \log(w_1 - g_1) + (1-\alpha_1)\log G.$

$$\frac{\partial U_1}{\partial g_1} = \frac{-\alpha_1}{w_1 - g_1} + \frac{(1-\alpha_1)}{g_1+g_2} = 0.$$

$$\frac{\alpha_1}{w_1 - g_1} = \frac{1-\alpha_1}{g_1+g_2}$$

$\alpha_1(g_1+g_2) = (1-\alpha_1)(w_1 - g_1)$

$\cancel{\alpha_1 g_1} + \alpha_2 g_2 = w_1 - g_1 - \alpha_1 w_1 + \alpha_1 g_1$

$g_1 = w_1 - \alpha_1 w_1 - \alpha_2 g_2.$

$= w_1(1-\alpha_1) - \alpha_2 g_2$

$g_1^* = w_1 - \alpha_1 w_1 - \alpha_2 g_2.$

$g_1^* = w_1(1-\alpha_1) - \alpha_2 g_2.$

$g_2 \uparrow$

From the reaction function we can see, optimal contribution of ~~the~~ individual 1 (when public good is privately provided)

depends positively on the other individuals contribution as well as their preference of the public good as g_2 is, g_1 is. The income in g is determined by the valuation of individual 2's public good ie α_2.

✓ (4)

Yes, each individuals contribution is dependent on their own income level. Higher their income level greater their contribution to the public good.

(d) From the previous part, we obtain the following reaction function.

$$g_1^* = w_1(1-\alpha_1) - \alpha_2 g_2$$

By symmetry

$$g_2^* = w_2(1-\alpha_2) - \alpha_1 g_1$$

To calculate N.E.

$$g_1^* = w_1(1-\alpha_1) - \alpha_2(w_2(1-\alpha_2) - \alpha_1 g_1)$$
$$= w_1(1-\alpha_1) - \alpha_2(w_2 - w_2\alpha_2 - \alpha_1 g_1)$$
$$= w_1(1-\alpha_1) - \alpha_2 w_2 + w_2\alpha_2^2 + \alpha_1\alpha_2 g_1$$
$$g_1^* = w_1 - w_1\alpha_1 - \alpha_2 w_2 + w_2\alpha_2^2 + \alpha_1\alpha_2 g_1$$
$$g_1 - \alpha_1\alpha_2 g_1 = w_1 - w_2\alpha_1 - \alpha_2 w_2 + w_2\alpha_2^2$$
$$g_1(1-\alpha_1\alpha_2) = w_1 - w_2\alpha_1 - \alpha_2 w_2 + w_2\alpha_2^2$$

(5)

$$g_1 = \frac{w_1 - w_2\alpha_1 - \alpha_2 w_2 + w_2\alpha_2^2}{(1-\alpha_1\alpha_2)}$$

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from reaction functions

$\frac{\partial g_1^*}{\partial \alpha_1} = g_1 = w_1(1-\alpha_1) - \alpha_2 g_2$

$g_2 = w_2(1-\alpha_2) - \alpha_1 g_1$

$\frac{\partial g_1^*}{\partial w_1} = \frac{1}{1-\alpha_1\alpha_2} > 0 \qquad \alpha_1\alpha_2 \leq 1$

This means that in NE, the opt contribution of individual 1 depends positively on his/her level of income. Higher the income, the more he/she is willing to contribute towards the public good when it is provided.

$\frac{\partial g_1^*}{\partial w_2} = \frac{-\alpha_1}{(1-\alpha_1\alpha_2)} - \frac{\alpha_2}{(1-\alpha_1\alpha_2)} + \frac{2\alpha_2 w_2}{(1-\alpha_1\alpha_2)}$

$= \frac{2\alpha_2 w_2 - (\alpha_1+\alpha_2)}{(1-\alpha_1\alpha_2)}$

from g_1^*:

$g_1 = w_1(1-\alpha_1) - \alpha_2\left[w_2(1-\alpha_2) - \alpha_1 g_1\right]$

$g_1 = w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2) + \alpha_1\alpha_2 g_1$

$g_1(1-\alpha_1\alpha_2) = w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2)$

$g_1 = \frac{w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2)}{1-\alpha_1\alpha_2}$

$g^* + g_2^* = G_{NE}$

$g_1^* = \frac{w_1 - w_2\alpha_1 - \alpha_1 w_2 + w_2\alpha_2^2}{(1-\alpha_1\alpha_2)} = w_1(1-\alpha_1) - \alpha_2 w_2$

$g_2^* = w_2 \frac{- w_1\alpha_2 - \alpha_1 w_1 + w_1\alpha_1^2}{(1-\alpha_1\alpha_2)}$

$G_{NE} = w_1 - w_2\alpha_1 - \alpha_1 w_2 + w_2\alpha_2^2 + w_2 - w_1\alpha_2 - \alpha_1 w_1 + w_1\alpha_1^2$

$= (w_1+w_2) - w_2(\alpha_1+\alpha_2) - w_1(\alpha_1+\alpha_2) + w_1\alpha_1^2 + w_2\alpha_2^2$

$= (w_1+w_2) - (w_1+w_2)(\alpha_1+\alpha_2) + w_1\alpha_1^2 + w_2\alpha_2^2$

$= (w_1+w_2)(1-\alpha_1+\alpha_2) + w_1\alpha_1^2 + w_2\alpha_2^2$

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from reaction functions

$\frac{\partial g_1^*}{\partial \alpha_1} = g_1 = w_1(1-\alpha_1) - \alpha_2 g_2$

$g_2 = w_2(1-\alpha_2) - \alpha_1 g_1$

$\frac{\partial g_1^*}{\partial w_1} = \frac{1}{1-\alpha_1\alpha_2} > 0 \qquad \alpha_1\alpha_2 \leq 1$

This means that in NE, the opt contribution of individual 1 depends positively on his/her level of income. Higher the income, the more he/she is willing to contribute towards the public good when it is provided.

$\frac{\partial g_1^*}{\partial w_2} = \frac{-\alpha_1}{(1-\alpha_1\alpha_2)} - \frac{\alpha_2}{(1-\alpha_1\alpha_2)} + \frac{2\alpha_2 w_2}{(1-\alpha_1\alpha_2)}$

$= \frac{2\alpha_2 w_2 - (\alpha_1+\alpha_2)}{(1-\alpha_1\alpha_2)}$

from g_1^*:

$g_1 = w_1(1-\alpha_1) - \alpha_2\left[w_2(1-\alpha_2) - \alpha_1 g_1\right]$

$g_1 = w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2) + \alpha_1\alpha_2 g_1$

$g_1(1-\alpha_1\alpha_2) = w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2)$

$g_1 = \frac{w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2)}{1-\alpha_1\alpha_2}$

$g^* + g_2^* = G_{NE}$

$g_1^* = \frac{w_1 - w_2\alpha_1 - \alpha_1 w_2 + w_2\alpha_2^2}{(1-\alpha_1\alpha_2)} = w_1(1-\alpha_1) - \alpha_2 w_2$

$g_2^* = w_2 \frac{- w_1\alpha_2 - \alpha_1 w_1 + w_1\alpha_1^2}{(1-\alpha_1\alpha_2)}$

$G_{NE} = w_1 - w_2\alpha_1 - \alpha_1 w_2 + w_2\alpha_2^2 + w_2 - w_1\alpha_2 - \alpha_1 w_1 + w_1\alpha_1^2$

$= (w_1+w_2) - w_2(\alpha_1+\alpha_2) - w_1(\alpha_1+\alpha_2) + w_1\alpha_1^2 + w_2\alpha_2^2$

$= (w_1+w_2) - (w_1+w_2)(\alpha_1+\alpha_2) + w_1\alpha_1^2 + w_2\alpha_2^2$

$= (w_1+w_2)(1-\alpha_1+\alpha_2) + w_1\alpha_1^2 + w_2\alpha_2^2$

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Question 1

(a). Firm 1

Given that there is no regulation, the damage function is not included.

$\pi_1 = (3-2)y_1$

$= 1y_1$

$= y_1$

$y_1 = 200$ units

Pollution $= 0.01y_1^2 = 400$ units

Firm 2, similarly in the absence of regulation

$\pi_2 = (3-2)y_2$

$\pi_2 = y_2$

At their max capacity = 200, they produce 200 units of oil.

pollution = 100 units

$\left(0.01\frac{y_2^2}{2}\right) = \frac{1}{100} \times 200^2$

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Total amount of smog produced as an outcome of competitive behaviour = 300 units (200 + 100)

(b) When externalities are internalized,

Max.

$\pi_1 = (3-2)y_1 - 0.01y_1^2$

$\frac{\partial \pi_1}{\partial y_1} = 1 - 2(0.01)y_1 = 0.$

$1 - 0.02y_1 = 0$

$1 = 0.02y_1$

$y_1^* = \frac{1}{0.02} = \frac{1}{\frac{2}{100}} = \frac{100}{2} = 50$ units

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similarly

$\pi_2 = (3-2)y_2 - \frac{(0.01)}{2} y_2^2$

for profit maximization

$\frac{\partial \pi_2}{\partial y_2} = 1 - \frac{2(0.01)}{2} y_2$

$= 1 - \frac{1}{100} y_2 = 0$

$1 = \frac{1}{100} y_2$

$y_2^* = 100$

Inorder to maximise profits firm$_2$ produces 100units

∴ The socially optimal level of smog

$= 0.01 y_1^{*2} + \frac{0.01 y_2^{*2}}{2}$

$= 0.01 \left(y_1^{*2} + \frac{y_2^2}{2} \right)$

$= \frac{1}{100} \left(50^2 + \frac{100^2}{2} \right)$

$= \frac{1}{100} \left(2500 + \frac{10000}{2} \right)$

$= \frac{1}{100} (2500 + 5000)$

$= \frac{7500}{100}$

$= 75$ units

V. good

10

The socially optimal outcome is lower than the outcome obtained

when competitive behaviour is permitted (in the absence of regulation). As the pollution is a negative externality, the market tends to overproduce.

✓ (4)

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for NoE to be socially beneficial.

$G_{NE}(2-\alpha_1-\alpha_2) \geq G_{NE}$.

$G_{NE}(1-\alpha_1-\alpha_2) \geq 0$.

$1-(\alpha_1+\alpha_2) \geq 0$

utility derived from G_{NE} > cost incurred in corruption G_{NE}

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CONTINUATION TEST ANSWER BOOK

Name: Sweta Patra

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Course Code: EC0-354 Course Title: Public economics.

Date: 06/10/16 Signature of Invigilator:

This continuation book will be evaluated only if all the above entries are complete. Write on all pages.

Reaction function of individual.

Q1 (d) $g^* = w_1(1-\alpha_1) - \alpha_2 g_2$

By symmetry

$g_2' = w_2(1-\alpha_2) - \alpha_1 g_1$

To calculate N.E provision of g_1^* & g_2^*

$g_1 = w_1(1-\alpha_1) - \alpha_2[w_2(1-\alpha_2) - \alpha_1 g_1]$

$g_1 = w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2) + \alpha_1\alpha_2 g_1$

$g_1(1-\alpha_1\alpha_2) = w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2)$

$g_1^* = \dfrac{w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2)}{1-\alpha_1\alpha_2}$

Typo !!

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$g_i = \dfrac{w_i(1-\alpha_i) - \alpha_j w_j(1-\alpha_j)}{1-\alpha_i\alpha_j}$

Agent's contribution positively depends on his/her income.

As $\dfrac{\partial g_i}{\partial w_i} = \dfrac{(1-\alpha_i)}{(1-\alpha_i\alpha_j)} > 0$ $\alpha_i\alpha_j \geq 0$.

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$$\frac{\partial g_i}{\partial w_j} = \frac{\alpha_j(1-\alpha_j)}{(1-\alpha_i\alpha_j)} < 0$$

Therefore, agent i's contribution ~~positively~~ negatively depends on the other agents income. If the other agents income is high, he will value the public good more thus agent 1 can free ride on the public good contribution of agent 2.

$$\cancel{\frac{\partial g_i}{\partial \alpha_i} = \frac{\partial g_i}{\partial \alpha_i} = -w_i(1-\alpha_i\alpha_j)^{-1} + -1(1-\alpha_i\alpha_j)^{-2} - w_i}$$

$$=$$

$$\frac{\partial g_i}{\partial \alpha_i} = -w_i(1-\alpha_i\alpha_j)$$

$$= -w_i(1-\alpha_i\alpha_j)^{-1} + -1(1-\alpha_i\alpha_j)^{-2} w_i(1-\alpha_i)$$

$$= w_i(1-\alpha_i\alpha_j)^{-1}\left[-1 + (1-\alpha_i\alpha_j)(1-\alpha_i)\right]$$

Higher agent 2's valuation of public good, ~~however~~ lower agent 1's contribution towards the public good
Higher agent 1's contribution, higher his/her contribution towards the public good.

(e) $G_{NE} = g_1^* + g_2^*$

$$= \frac{w_1(1-\alpha_1) - \alpha_2 w_2(1-\alpha_2) + w_2(1-\alpha_2) - \alpha_1 w_1(1-\alpha_1)}{1-\alpha_1\alpha_2}$$

$$= \frac{(1-\alpha_2)(w_2 - \alpha_1 w_2) + (1-\alpha_1)(w_1 - \alpha_1 w_1)}{1-\alpha_1\alpha_2}$$

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$$w_2 \frac{(1-\alpha_2)(1-\alpha_2) + (1-\alpha_1)w_1(1-\alpha_1)}{(1-\alpha_1\alpha_2)}$$

$$w_2 \frac{(1-\alpha_1)^2 + w_1(1-\alpha_1)^2}{(1-\alpha_1\alpha_2)} = G_{NE}.$$

$$G^0 = (1-\alpha)(w_1+w_2)$$

G_{NE} (when $\alpha_1 = \alpha_2 = \alpha$)

$$G_{NE} = \frac{w_2(1-\alpha)^2 + w_1(1-\alpha)^2}{(1-\alpha^2)}$$

$$= \frac{(w_1+w_2)(1-\alpha)^2}{(1-\alpha^2)}$$

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$$= \cancel{(w_1+w_2)(1-\alpha)}$$

We know that $G_{NE} < G_0$ as the public good tends to be under provided when privately provided

since $\alpha \leq 1$

$$\frac{(w_1+w_2)(1-\alpha)^2}{(1-\alpha^2)} < (w_1+w_2)(1-\alpha)$$

$$\frac{(w_1+w_2)(1+\alpha^2-2\alpha)}{} < (w_1+w_2)(1-\alpha)(1-\alpha^2)$$

$$(w_1+w_2)(1-\alpha)^2 < (w_1+w_2)(1-\alpha)(1-\alpha)(1+\alpha)$$

$$1 < (w_1+w_2)(1+\alpha)$$

$$(w_1+w_2) > 1 \qquad (1+\alpha) > 1$$

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(f) In the N E allocation.

$u = \alpha \log x_1 + (1-\alpha) \log G$

Utility derived from public good by agent i

$= (1-\alpha_i)(g_1+g_2) = (1-\alpha_i) G$

$= G_{NE} = \frac{(w_1+w_2)(1-\alpha)^2}{(1-\alpha)}$

$U(g_1) = (1-\alpha_1)(g_1+g_2)$
$U(g_2) = (1-\alpha_2)(g_1+g_2)$

Utility derived by both agents of consumption of public good $= (1-\alpha_1) G^*_{NE} + (1-\alpha_2) G^*_{NE}$

$= G^*(1-\alpha_1 + 1-\alpha_2)$

$= G^*_{NE}(2-\alpha_1-\alpha_2)$

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Cost = share of income spent on the public good

$(1-\alpha_1) = \frac{g_1}{w_1}$ $(1-\alpha_2) = \frac{g_2}{w_2}$.

Cost: $(1-\alpha_1) + (1-\alpha_2)$

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or

$= g_1^* + g_2^* = G_{NE}$